Literature Review (Short Version)

Existing communication apps compared with SACP

> **About SACP.** SACP stands for Smart Academic Communication Platform. It helps a university send the right message to the right people at the right time, using email, SMS, push and in-app messages. It has three user roles: **Admin**, **Faculty Member** and **Student**. Each table below shows what an existing app offers, what it is missing, and how SACP fills that gap.

1. Slack

Team chat app

Slack is a work chat app built around channels, direct messages and file sharing. It also has a simple tool for building small automations. It is made for office teams, not for schools.

Area	What Slack provides	What it lacks	What SACP provides and how it fills the gap
Channels and reach	Public and private channels, direct messages, threads, file sharing and quick calls.	Works only inside Slack. It cannot send email, SMS or push alerts to people who are not using it.	Sends the same notice by email, SMS, push and inside the app, so students are reached wherever they are.
Audience	Channels that people join or are added to.	Channels are not linked to class lists. Someone has to add members by hand.	Builds groups straight from course and student records, for example "all students in CS101".
Automation	A simple workflow builder (paid) for forms and basic actions.	No academic triggers such as an exam being posted. Conditions are basic.	Rules react to academic events and can check conditions, for example low grade and unpaid fee, then send an alert.
Templates and scheduling	Can schedule a single message and set personal reminders.	No reusable message templates. No scheduled bulk announcements.	Reusable templates with fields like student name and course. Notices can be sent once or repeated on a schedule.
Incoming messages	Replies in channels and direct messages. No smart sorting.	Every message looks the same, so urgent questions get buried.	AI sorts student questions by urgency so faculty see the important ones first.
Reports and roles	Basic usage stats. Generic roles such as owner and member.	No tracking of who opened a message. No faculty or student roles.	Tracks opens and clicks. Uses three clear roles: Admin, Faculty Member and Student. Students cannot send announcements.

> **Summary: Slack vs SACP.** Slack is good for everyday team chat, but it does not know anything about courses, exams or grades. SACP keeps the organised-channel idea but adds ready-made class groups, templates, scheduling and alerts outside the app.

2. Microsoft Teams

Chat, meetings and collaboration suite

Teams is part of Microsoft 365. It combines chat, video meetings, calls and file sharing, and connects closely with Word, Outlook and other Microsoft apps.

Area	What Microsoft Teams provides	What it lacks	What SACP provides and how it fills the gap
Channels and reach	Team channels, group chat, calls and meetings.	Stays inside the Microsoft world. No SMS or push alerts to students who are not on Teams.	Reaches students by email, SMS and push even if they never open a chat app.
Audience	Teams built from invited members. Guests can join.	Not linked to courses. Members must be managed manually or synced separately.	Picks audiences automatically from course and section lists.
Automation	Works with Power Automate for custom flows.	No built-in easy builder. Academic triggers need custom setup.	Built-in rules, for example "when an exam is graded, notify students by email and in the app".
Templates and scheduling	Announcement posts and calendar invites.	No message template library. Channel posts cannot be scheduled or repeated easily.	Reusable templates and scheduled or repeating announcements, such as a drop-date reminder next Monday at 9 AM.
Targeting and AI	Can mention a whole team or tag a group. Summaries from Copilot.	Cannot target students by result or status. No sorting of student questions.	Targets by course, status or condition, and AI flags urgent student questions.
Roles	Owner and member, plus guests.	No education-specific roles.	Admin, Faculty Member and Student roles with clear limits on who can send or change what.

> **Summary: Microsoft Teams vs SACP.** Teams is strong for virtual classes and meetings, but sending timed, rule-based alerts to the right students needs extra custom work. SACP does this out of the box and over more channels.

3. Twilio / SendGrid

Messaging services for developers

Twilio sends SMS, WhatsApp and other messages through code. SendGrid, its email service, sends email at large scale. They are building blocks for developers, not ready-made apps for staff.

Area	What Twilio provides	What it lacks	What SACP provides and how it fills the gap
Channels	SMS, WhatsApp and email through one set of tools. Reliable at large scale.	No screen for everyday users and no chat. No push alerts or in-app messages.	Gives staff a simple screen to send messages, plus push alerts and an in-app chat. It can use services like these to deliver messages.
Audience	Contact lists that can be imported and grouped.	No idea of courses or students. Lists must be prepared by hand.	Creates lists from academic records, for example all students registered for an exam.
Templates	Dynamic email templates. WhatsApp needs pre-approved templates.	Each channel has its own rules and limits.	Write a message once and SACP fits it to each channel.
Scheduling and automation	Scheduling and flows are possible with code.	No ready scheduler or easy rule builder. A developer must write the logic.	Built-in scheduler and rule builder that non-technical staff can use.
Replies	Replies can be received and passed to code.	No conversation screen. Replies must be handled by custom code.	Replies show up as a conversation in the SACP app and go through AI sorting.
Cost and tracking	Pay per message. Basic delivery logs.	Pricing can be complex. Little insight into who actually read a message.	Tracks delivery and reads, and can choose cheaper channels such as push when possible.

> **Summary: Twilio vs SACP.** Twilio and SendGrid are the pipes that carry messages. They do not decide who should get what or when. SACP adds the screens, student data, rules and sorting on top, so staff need no coding.

4. Microsoft Outlook

Email and calendar app

Outlook handles email, calendar, contacts and tasks. It is great for personal email and meeting invites, but it knows nothing about courses.

Area	What Microsoft Outlook provides	What it lacks	What SACP provides and how it fills the gap
Channels	Email, calendar, contacts and tasks.	Email and calendar only. No SMS or in-app chat.	Adds SMS, push and in-app messages so urgent news is not missed.
Audience	Contact groups and distribution lists.	Lists are made by hand and do not follow course enrolment.	Builds email lists from course rosters automatically.
Templates and scheduling	Quick Parts, simple templates and delayed sending for single emails.	No placeholders and no easy bulk scheduling to a whole list.	Templates with fields, and scheduled bulk notices such as fee reminders.
Targeting	Manual choice of recipients.	Cannot pick students by grade, year or similar details.	Faculty choose by class or rule, for example "grade below 40%".
Incoming mail and tracking	Focused Inbox sorts mail. Optional read receipts.	Student questions mix with all other mail. No tracking of who read a notice.	A separate question view sorted by urgency, with open and click tracking.

> **Summary: Microsoft Outlook vs SACP.** Outlook is a solid email tool, but sending targeted, scheduled, multi-channel notices to a class is mostly manual. SACP covers that and adds SMS, push and AI sorting.

5. Google Classroom

Learning platform for classes

Classroom lets teachers create classes, share work, grade and post announcements in a class stream. It works with Google Drive, Calendar, Meet and Gmail.

Area	What Google Classroom provides	What it lacks	What SACP provides and how it fills the gap
Course management	Classes with students, assignments, materials and grades.	This is its strength, but it is not a messaging tool.	SACP does not replace Classroom. It handles the communication around the classes.
Announcements	Class stream posts, with optional email summaries and app alerts.	Posts stay in Classroom. No SMS. No notices across all classes at once.	Sends the same announcement by email, SMS or push, and college-wide notices when needed.
Reminders and automation	Due dates shown to students. Everything else is done by hand.	No rule-based alerts, for example when many students are late.	Automatic reminders such as "Assignment due in 3 days", sent without reposting.
Templates and questions	Ad-hoc posts. Comment threads if turned on.	No templates and no sorting of student questions.	Templates for common notices and AI sorting of questions by urgency.
Reports and roles	Gradebook. Teacher, co-teacher, student and guardian roles.	No stats on who read announcements. No roles above a single class.	Message analytics and an Admin role that works across all classes.

> **Summary: Google Classroom vs SACP.** Classroom is good for running a class but limited for sending flexible alerts. SACP works next to it, sending reminders and notices over more channels with less manual effort.

6. Discord

Community chat app

Discord is a chat app for communities, with text and voice channels, threads and detailed permissions. It began with gaming but is now used by many groups.

Area	What Discord provides	What it lacks	What SACP provides and how it fills the gap
Channels	Servers with topic channels, voice rooms and forum-style threads.	People join by invite. Channels are not tied to real courses.	Organised groups like Discord, but created automatically from real classes.
Roles	Flexible roles with fine permissions.	Roles are set by hand and do not match school roles.	Roles come from institution data: Admin, Faculty Member, Student. Only faculty and admins send broadcasts.
Notices and scheduling	Scheduled events and bots that can post.	No templates. No recurring notices without bots.	Templates and built-in scheduling for any notice, not only events.
Reach	Messages inside Discord only.	Students must be on Discord to see anything.	Email, SMS and push reach students even when they are offline.
Replies and tracking	Threads and bot replies. Very basic stats.	All replies look equal. No info on who saw a message.	Urgent questions rise to the top, and faculty can see who opened a notice.

> **Summary: Discord vs SACP.** Discord is great for open, friendly chat but has no link to school systems and no way to send alerts outside the app. SACP is more controlled and driven by real student data.

7. WhatsApp

Mobile messaging app

WhatsApp is the most common phone chat app. It offers one-to-one chats, group chats, broadcast lists and, through its business service, template messages.

Area	What WhatsApp provides	What it lacks	What SACP provides and how it fills the gap
Chats	Direct chats, groups of up to 1,024 people, calls and broadcast lists.	Phone-based. Group chats are crowded and have no real threads. Broadcast lists are small.	Keeps questions in tidy threads inside SACP, and can use WhatsApp as one of several channels.
Audience	Contacts from the phone.	Faculty must save every student's number. No class lists.	Uses student phone numbers already on record, not anyone's phonebook.
Templates and scheduling	Business templates, which WhatsApp must approve.	Limited wording. No scheduling in the normal app.	Manages templates in one place and sends WhatsApp messages at chosen times.
Two-way chat	Fully two-way.	No way to tell which messages matter most. No read stats for senders.	Replies come into SACP, where AI marks which ones need attention first.
Coverage	Strong privacy through encryption.	Not everyone uses it.	Also offers email, SMS and push for those who do not.

> **Summary: WhatsApp vs SACP.** WhatsApp reaches almost everyone but has no automation or organisation for academic use. SACP can include it as one channel while adding scheduling, rules and sorting.

8. Mailchimp

Email marketing platform

Mailchimp helps businesses design newsletters, group their audience and run automatic email series. It is built for marketing, not for schools.

Area	What Mailchimp provides	What it lacks	What SACP provides and how it fills the gap
Channels	Mainly email, with limited SMS.	No chat, no push alerts and no app of its own.	Treats email, SMS and push as equal channels.
Templates	Drag-and-drop email builder and many designs.	Email only. Not built for notices filled with student details.	Templates for academic messages that fill in student and course details.
Automation	Automatic series, such as welcome emails and triggers from customer actions.	Triggers come from shopping and website activity, not from grades or exams.	Automation reacts to academic events, such as a missed submission.
Groups and timing	Detailed audience groups. Best-send-time tool and A/B tests.	Groups are built from its own lists, not school data. Timing is based on marketing habits.	Groups come from student data. Timing follows exam dates and term calendars.
Replies and reports	Open and click reports. Replies go to one email address.	No conversation view. Reports measure sales-style success.	Replies become threads in the app. Reports show things like how many in a class opened the exam notice.

> **Summary: Mailchimp vs SACP.** Mailchimp has useful ideas such as templates, automation and reports, but they serve selling and marketing. SACP uses similar ideas to send needed information to students and courses.

9. Rocket.Chat

Open-source team chat

Rocket.Chat is like Slack but open source. Organisations can run it on their own servers and change it as they like.

Area	What Rocket.Chat provides	What it lacks	What SACP provides and how it fills the gap
Chat and control	Channels, calls, file sharing, and self-hosting with strong security.	Channels and members still need to be set up by hand. Changes need developers.	Uses the same channel idea but fills groups from academic data.
Apps and bots	A marketplace of apps and bots.	No ready connections to school systems.	Comes with academic connections already planned.
Broadcasts and templates	Posting in channels and direct messages only.	No broadcast tool and no message templates.	Templates, scheduling and broadcasts by email and SMS.
Alerts and AI	Push alerts for chat messages only.	No alerts tied to outside events. No AI sorting.	Alerts for any academic event, over many channels, with AI sorting of replies.
Reports and roles	Basic usage stats. Admin, moderator and user roles.	No message reach reports. Roles are not school-specific.	Message analytics and Admin, Faculty Member and Student roles.

> **Summary: Rocket.Chat vs SACP.** Rocket.Chat gives a safe, flexible place to chat, but it stays a chat tool. SACP sits one step higher as a communication system built around academic events.

10. Moodle

Open-source learning platform

Moodle is a full learning platform with courses, assignments, quizzes, forums and grades. It also sends notifications by email and app, and admins can post site-wide announcements.

Area	What Moodle provides	What it lacks	What SACP provides and how it fills the gap
Courses	Full course tools and grading.	Not focused on messaging, and many teachers do not use the forums.	Does not replace Moodle. It listens for events from it and handles the messages.
Notifications	Alerts for due dates, grades and forum posts by email or app.	Mostly reactive and decided by settings. Push needs the mobile app.	Smart rules, such as overdue work plus low grade sends an alert. SMS and push work without the Moodle app.
Message wording	Fixed default texts.	Teachers cannot easily change the wording or reuse their own.	Template manager so teachers can write and reuse their own messages.
Targeting	Course members and groups inside a course.	Hard to target across courses, such as all failing students.	Targets by major, year or performance across courses.
Questions and reports	Forums for questions. Course and grade reports.	Questions have no priority. No report on who read a notice.	Questions sorted by urgency, with open and click tracking.

> **Summary: Moodle vs SACP.** Moodle is strong at teaching and grading but basic at flexible communication. SACP uses its data to send timely, targeted messages across several channels.

11. Spark Mail

Smart email app

Spark is an email app for phones and computers. It has a smart inbox, shared inboxes, team comments, templates and AI help for writing and summaries.

Area	What Spark Mail provides	What it lacks	What SACP provides and how it fills the gap
Email handling	Smart inbox, snooze and quick actions.	Email only.	Adds SMS and push, and borrows the idea of flagging important messages.
Templates and sending	Personal and shared templates. Send later for single emails.	Not made for bulk or event-based notices.	Templates and scheduling that work for large groups and automatic notices.
Teamwork	Shared inboxes and assigned emails.	Still email-based, with no chat.	Many faculty and staff can use SACP at the same time in one system.
AI	Summaries and draft replies for personal use.	Works on one person's mail, not group communication.	AI sorts messages by what they say, not only who sent them.
Tracking	Basic read receipts.	No stats for broadcasts.	Server-side tracking of delivery, opens and clicks.

> **Summary: Spark Mail vs SACP.** Spark makes personal email smarter, but it stays inside email. SACP offers broadcasting, rules and extra channels for institutions.

12. Gmail

Email service from Google

Gmail is the most used email service. It offers labels, filters, canned replies, scheduled sending and tabs such as Primary and Promotions.

Area	What Gmail provides	What it lacks	What SACP provides and how it fills the gap
Channels	Email, with Calendar and Meet.	No SMS or push alerts.	Adds the missing channels.
Audience	Contact groups, labels and filters.	No student lists. Class groups must be kept by hand.	Student lists update automatically from records.
Templates and scheduling	Canned responses and Schedule Send.	Plain text only and one email at a time. No bulk scheduling screen.	Shared templates with fields and bulk scheduling to large groups.
Rules	Filters that act on incoming mail.	Cannot react to outside events such as an exam being posted.	Rules start from academic events, not from incoming mail.
Visibility and tracking	Smart tabs and priority sorting. Almost no tracking of sent mail.	Class emails may land in Promotions and be missed. Cannot see who opened a message.	Important notices go out by SMS or push too, with open and click tracking.

> **Summary: Gmail vs SACP.** Gmail is easy and familiar, but only for email and with no school context. SACP works on top of any email service and adds orchestration across channels.